

Fluid Dynamics- Numerical Techniques

MATH5453M FEM Numerical exercise 3, 2025

Student: Nipon Roy (201480731)

Given the Poisson system of equations,

$$-\nabla^2 u = f \quad \text{on } (x, y) \in [0, 1]^2 \dots \dots \dots (1a)$$

$$f(x, y) = 2\pi^2 \sin(\pi x) \cos(\pi y) \dots \dots \dots (1b)$$

$$u(0, y) = u(1, y) = 0 \dots \dots \dots (1c)$$

$$\partial_y u(x, y)|_{y=0} = \partial_y u(x, y)|_{y=1} = 0 \dots \dots \dots (1d)$$

The exact solution given, $u_e(x, y) = \sin(\pi x) \cos(\pi y) \dots \dots \dots (2)$

To check that the eq. (2) is the exact solution for the system of equations in (1), we must prove that the LHS equals to the RHS of the eq. (1a) and the boundary conditions are also satisfied.

Let us substitute this exact solution into the eq. (1a), first for the Laplacian the 2nd derivatives need to be evaluated,

$$\frac{\partial^2 u(x, y)}{\partial x^2} = -\pi^2 \sin(\pi x) \cos(\pi y) \dots \dots \dots (3a)$$

$$\frac{\partial^2 u(x, y)}{\partial y^2} = -\pi^2 \sin(\pi x) \cos(\pi y) \dots \dots \dots (3b)$$

Now from eq. (1),
$$-\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 2\pi^2 \sin(\pi x) \cos(\pi y)$$

$$-(-2\pi^2 \sin(\pi x) \cos(\pi y)) = 2\pi^2 \sin(\pi x) \cos(\pi y)$$

$$LHS = RHS \text{ (checked).}$$

Checking the Dirichlet and Neumann boundary conditions, respectively:

$$u(0, y) = \sin(\pi \cdot 0) \cos(\pi y) = 0$$

$$u(1, y) = \sin(\pi \cdot 1) \cos(\pi y) = 0$$

$$\partial_y u(x, y)|_{y=0} = -\pi \sin(\pi x) \sin(\pi \cdot 0) = 0$$

$$\partial_y u(x, y)|_{y=1} = -\pi \sin(\pi x) \sin(\pi \cdot 1) = 0$$

All the given conditions are satisfied. Therefore, equation (2) is the verified exact solution.

Now moving to answering the questions:

Question 1

The Ritz-Galerkin principle for the above Poisson system:

According to the Ritz-Galerkin principle, the minimiser of an energy functional is the possible solution to the given Poisson system of equations. Therefore, an energy functional $I(u)$ is defined over 2D domain $\Omega=[0,1]^2$. The true solution u can minimize this functional:

$$I(u) = \int_{\Omega} \left[\frac{1}{2} |\nabla u|^2 - fu \right] d\Omega \dots \dots \dots (1)$$

To find the minimum, we take the variation δI and set it equal to zero ($\delta I = 0$). It must also maintain the Dirichlet boundary conditions: $\delta u(0, y) = \delta u(1, y) = 0$.

Let u vary by a small δu and taking the first variation of the $I(u)$ by setting,

$$\delta I = \frac{d}{d\varepsilon} I[(u + \varepsilon \delta u)] \Big|_{\varepsilon=0} = 0$$

$$\delta I = \int_{\Omega} [\nabla u \cdot \nabla(\delta u) - f \delta u] d\Omega = 0 \dots \dots \dots (2)$$

Now, applying integration by parts in 2D,

$$\int_{\Omega} [\nabla u \cdot \nabla(\delta u)] d\Omega = - \int_{\Omega} \nabla^2 u \delta u d\Omega + \int_{\partial\Omega} (\nabla u \cdot \hat{n}) \delta u d\Gamma \dots \dots \dots (3)$$

Substituting eq.(3) into eq.(2),

$$\int_{\Omega} ((-\nabla)^2 u - f) \delta u d\Omega + \int_{\partial\Omega} \frac{\partial u}{\partial n} \delta u d\Gamma = 0 \dots \dots \dots (4)$$

For any arbitrary variation δu , the inner terms of equation (4) must vanish independently. And on the Neumann boundary: $\frac{\partial u}{\partial n} = 0$.

$$-\nabla^2 u - f = 0$$

$$-\nabla^2 u = f \dots \dots \dots (5)$$

Thus, the equation (5) resulted from the variation method is the same Poisson system of equations.

The conditions that the variation must satisfy are

1. It must maintain the Dirichlet boundary conditions. It must be zero on the boundaries: $\delta u(0, y) = \delta u(1, y) = 0$.
2. It must be square integrable and must have square integrable first derivatives on the Dirichlet boundaries.

Weak formulation: the Galerkin method starts directly from the Poisson equation multiplied by the given test function, $\omega(x,y)$ and then integrating over the domain Ω .

$$\int_{\Omega} (-\nabla^2 u) \omega d\Omega = \int_{\Omega} f \omega d\Omega \dots \dots \dots (6)$$

$$\text{or, } \int_{\Omega} [\nabla u \cdot \nabla \omega] d\Omega - \int_{\Omega} \frac{\partial u}{\partial n} \omega d\Gamma = \int_{\Omega} f \omega d\Omega \dots \dots \dots (7)$$

Applying the boundary conditions, the final weak form becomes

$$\int_{\Omega} [\nabla u \cdot \nabla \omega] d\Omega = \int_{\Omega} f \omega d\Omega \dots \dots \dots (8)$$

Comparing the equation (8) with the equation (2),

$$\omega(x, y) = \delta u(x, y) \dots \dots \dots (9)$$

Question 2

FEM expansion:

Approximating the continuous solution $u(x, y)$ with a discrete finite element approximation $u_h(x, y)$ using a linear combination of global basis (shape) functions $\phi_j(x, y)$:

$$u(x, y) \approx u_h(x, y) = \sum_{j=1}^N u_j \phi_j(x, y) \dots \dots \dots (10),$$

where u_j is the coefficient of the basis functions and N is the total number of unconstrained nodes.

To build the discrete functional $I(u_h)$, substituting the FEM expansion directly into the continuous energy functional from eq. (1) for Discrete Ritz-Galerkin Principle:

$$I(u_h) = \int_{\Omega} \left[\frac{1}{2} \left(\nabla \left(\sum_{j=1}^N u_j \phi_j \right) \cdot \nabla \left(\sum_{i=1}^N u_i \phi_i \right) \right) - f \left(\sum_{j=1}^N u_j \phi_j \right) \right] d\Omega \dots \dots \dots (11)$$

Using linearity, creating an algebraic matrix equation:

$$I(\mathbf{u}) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N u_i u_j \left(\int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j d\Omega \right) - \sum_{j=1}^N u_j \left(\int_{\Omega} f \phi_j d\Omega \right) \dots \dots \dots (12)$$

In matrix-vector form, this algebraic principle becomes

$$I(\mathbf{u}) = \frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u} - \mathbf{u}^T \mathbf{F} \dots \dots \dots (13),$$

where the $\mathbf{u} = [u_1, u_2, u_3, \dots, u_N]^T$ is the vector of coefficients, and the $\mathbf{K}$ and $\mathbf{F}$ tensors can be defined as,

$$K_{ij} = \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j d\Omega$$

$$F_j = \int_{\Omega} f \phi_j d\Omega$$

For the discrete weak formulation, substituting u_h and $\omega_h = \phi_i$ into the continuous weak form in eq.(8) yields:

$$\int_{\Omega} \nabla \left(\sum_{j=1}^N u_j \phi_j \right) \cdot \nabla \phi_i d\Omega = \int_{\Omega} f \phi_i d\Omega \text{ for } i = 1, 2, \dots, N, \dots \dots \dots (14)$$

$$\sum_{j=1}^N u_j \left(\int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i d\Omega \right) = \int_{\Omega} f \phi_i d\Omega \text{ for } i = 1, 2, \dots, N, \dots \dots \dots (15)$$

In matrix vector notation, this is the familiar linear system:

$$K\mathbf{u} = \mathbf{F} \dots \dots \dots (16)$$

Differentiating $I(\mathbf{u})$ component by component with respect to an arbitrary node u_i for finding the extremum: $\frac{\partial I(\mathbf{u})}{\partial u_i} = 0, \text{ for } i = 1, 2, \dots, N.$

$$\frac{\partial I(\mathbf{u})}{\partial u_i} = \frac{\partial}{\partial u_i} \left[\frac{1}{2} \sum_{k=1}^N \sum_{j=1}^N u_k u_j k_{kj} - \sum_{j=1}^N u_j F_j \right] = 0 \dots \dots \dots (17)$$

$$\frac{\partial}{\partial u_i} \left[\frac{1}{2} \sum_{k=1}^N \sum_{j=1}^N u_k u_j k_{kj} \right] = \frac{\partial}{\partial u_i} \left[\sum_{j=1}^N u_j F_j \right] \dots \dots \dots (18)$$

$$\sum_{j=1}^N K_{ij} u_j = F_i \dots \dots \dots (19)$$

Substituting explicit integrals,

$$\sum_{j=1}^N u_j \left(\int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j d\Omega \right) = \int_{\Omega} f \phi_i d\Omega \dots \dots \dots (20)$$

Comparing this with the eq. (15) for the Discrete weak formulation, it is proved that minimizing the discrete Ritz functional is mathematically equivalent to solving the discrete Galerkin weak formulation.

Question 3

Local coordinate system and reference coordinates:

Considering the finite elements (mesh) in two dimensions, the domain is partitioned into a mixture of N_{el} quadrilateral elements such that the tessellation becomes

$$\tau_h = \left\{ K_k \mid \bigcup_{k=1}^{N_{el}} \overline{K_k} = \overline{\Omega} \text{ and } K_k \cap K_{k'} = \emptyset \text{ if } k \neq k', 1 \leq k, k' \leq N_{el} \dots \dots \dots (21) \right.$$

with $\overline{K_k}$ the closure of element K_k and each such element K_k has $N_n^k = N_n = 4$ nodes, which are numbered locally in counterclockwise fashion from 0 to $N_n - 1$.

To simplify the integration, every physical quadrilateral element is mapped from the global coordinates (x, y) to a standard reference space (ξ, η) . This reference element is a square spanning, $\xi \in [-1, 1]$ and $\eta \in [-1, 1]$. Using the isoparametric concept and the local bilinear shape functions $\hat{\phi}_a(\xi, \eta)$ to map the geometric coordinates (x, y) from the local reference element to the physical element:

$$\begin{aligned} x(\xi, \eta) &= \sum_{a=1}^4 x_a^e \hat{\phi}_a(\xi, \eta), \\ y(\xi, \eta) &= \sum_{a=1}^4 y_a^e \hat{\phi}_a(\xi, \eta) \end{aligned} \dots \dots \dots (22).$$

where (x_a^e, y_a^e) are the physical global coordinates of element's four vertices or nodes ($a=1,2,3,4$). Therefore, the local shape functions for these 4 nodes are

$$\begin{aligned} \hat{\phi}_1(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 - \eta) \\ \hat{\phi}_2(\xi, \eta) &= \frac{1}{4}(1 + \xi)(1 - \eta) \\ \hat{\phi}_3(\xi, \eta) &= \frac{1}{4}(1 + \xi)(1 + \eta) \\ \hat{\phi}_4(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 + \eta) \end{aligned} \dots \dots \dots (23).$$

To transfer the derivatives from the physical space $(\frac{\partial}{\partial x}, \frac{\partial}{\partial y})$ to the reference space $(\frac{\partial}{\partial \xi}, \frac{\partial}{\partial \eta})$ within the integrals, defining the Jacobian matrix [J]:

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \dots \dots \dots (24)$$

The physical spatial gradients change via the inverse Jacobian $[J]^{-1}$:

$$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = [J]^{-1} \begin{bmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{bmatrix} \dots \dots \dots (25)$$

The differential area transforms cleanly by the determinant of the Jacobian: $d\Omega = dx dy = \det([J]) d\xi d\eta$. Then the Jacobian matrix $[J_4]$ for the 4 elements:

$$[J_4(\xi, \eta)] = \frac{1}{4} \begin{bmatrix} (1-\eta)(x_2^e - x_1^e) + (1+\eta)(x_3^e - x_4^e) & (1-\eta)(y_2^e - y_1^e) + (1+\eta)(y_3^e - y_4^e) \\ (1-\xi)(x_4^e - x_1^e) + (1+\xi)(x_3^e - x_2^e) & (1-\xi)(y_4^e - y_1^e) + (1+\xi)(y_3^e - y_2^e) \end{bmatrix} \dots \dots (26)$$

Now the element stiffness matrix (K_{ab}^e) :

$$K_{ab}^e = \int_{-1}^1 \int_{-1}^1 (\nabla \hat{\phi}_a \cdot \nabla \hat{\phi}_b) \det([J]) d\xi d\eta \dots \dots \dots (27)$$

Here, ‘a’ corresponds to the row index (test function ϕ_i) and ‘b’ corresponds to the column index (trial solution ϕ_j) mapped locally $(a, b \in \{1, 2, 3, 4\})$. Element load vector (F_a^e) :

$$F_a^e = \int_{-1}^1 \int_{-1}^1 f(x(\xi, \eta), y(\xi, \eta)) \hat{\phi}_a(\xi, \eta) \det([J]) d\xi d\eta \dots \dots \dots (28)$$

Question 4

Figure 1 shows the provided exact analytical solution u_e for the 2D Poisson equation calculated on a discrete mesh grid size of 200 points on the domain $\Omega=[0,1]^2$. The solution is showing spatial distributions featuring a positive peak at the bottom centre and a symmetric negative region at the top centre. The uniformity along the left and right vertical edges represents the applied Dirichlet boundary conditions.

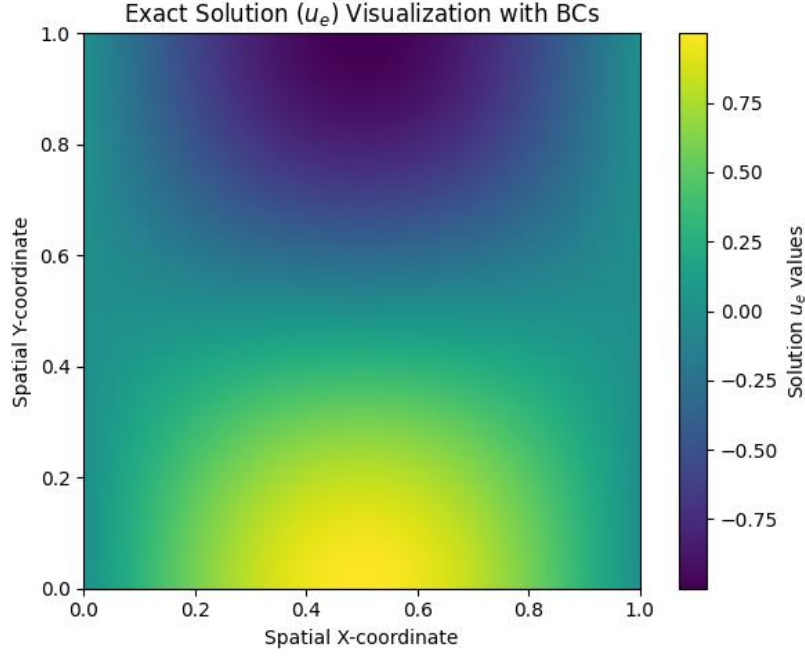


Figure 1: The exact (u_e) solution for the 2D Poisson equation.

Running the provided Firedrake script in Python using solution method 1 (the Discrete Galerkin Weak Formulation) and solution method 2 (Variational principle), the numerical solution u_h , is obtained on different mesh sizes ($n_x=n_y$) and on different polynomial orders using u1 and u2 solution methods. From table 1, L2 error between the u1 and u2 solution methods increases as the mesh resolution increases. This may not be a problem of the numerical methods, perhaps, it resulted from the numerical round off errors due to very high numbers of elements. Maximum absolute error $|u1-u_e|$ for solution method u1 against the exact solution decreases as the mesh resolution increases. The same trend is also true for the solution method u2.

Table 1: Error calculations for different resolutions (Δx) and a degree of polynomials $p=1$.

Items	$\Delta x = 0.0625$	$\Delta x = 0.03125$	$\Delta x = 0.015625$	$\Delta x = 0.0078125$
L2 norm between u1 and u2	9.8e-16	4.84e-15	2.69e-14	8.72e-14
Max absolute error $ u1-u_e $	3.2066e-03	8.0280e-04	2.0077e-04	5.0198e-05
Max absolute error $ u2-u_e $	3.2066e-03	8.0280e-04	2.0077e-04	5.0198e-05

Table 2 lists the L2 errors between u1 and u2 for different orders of polynomials and a mesh resolution of $\Delta x=0.0078125$ ($n_x=n_y=128$). The L2 errors between u1 and u2 show similar phenomena as in the table 1. However, the absolute errors for both u1 and u2 methods go down exponentially with the increase of the orders of the polynomials. The dramatic change takes place for going from $p=1$ to $p=2$. But the calculation time increases with increase in degrees of polynomials.

Table 2: Error calculated for different degrees of polynomials (p) and a mesh resolution of $\Delta x=0.0078125$ ($n_x=n_y=128$).

Items	P=1	P=2	P=3	P=4
L2 norm between u1 and u2	8.72e-14	2.35e-12	1.299e-12	1.207e-12
Max absolute error $ u_1 - u_e $	5.0198e-05	6.3772e-10	1.1346e-11	1.2282e-11
Max absolute error $ u_2 - u_e $	5.0198e-05	6.3040e-10	5.1912e-12	3.8676e-12

Figure 2 shows the numerical finite element solution u_h computed using quadrilateral grid for the 2D Poisson equation. Both solutions u_1 and u_2 are showing identical spatial distributions featuring a positive peak at the bottom centre and a symmetric negative region at the top centre. The uniformity along the left and right vertical edges represents the Dirichlet boundary conditions. This is similar with the exact solution u_e .

For the mesh resolution: $\Delta x = 0.0625$ and $p=1$. The L2 error for u_1 is 0.001593, and for u_2 is 0.00159301. Numerically it is quite low difference.

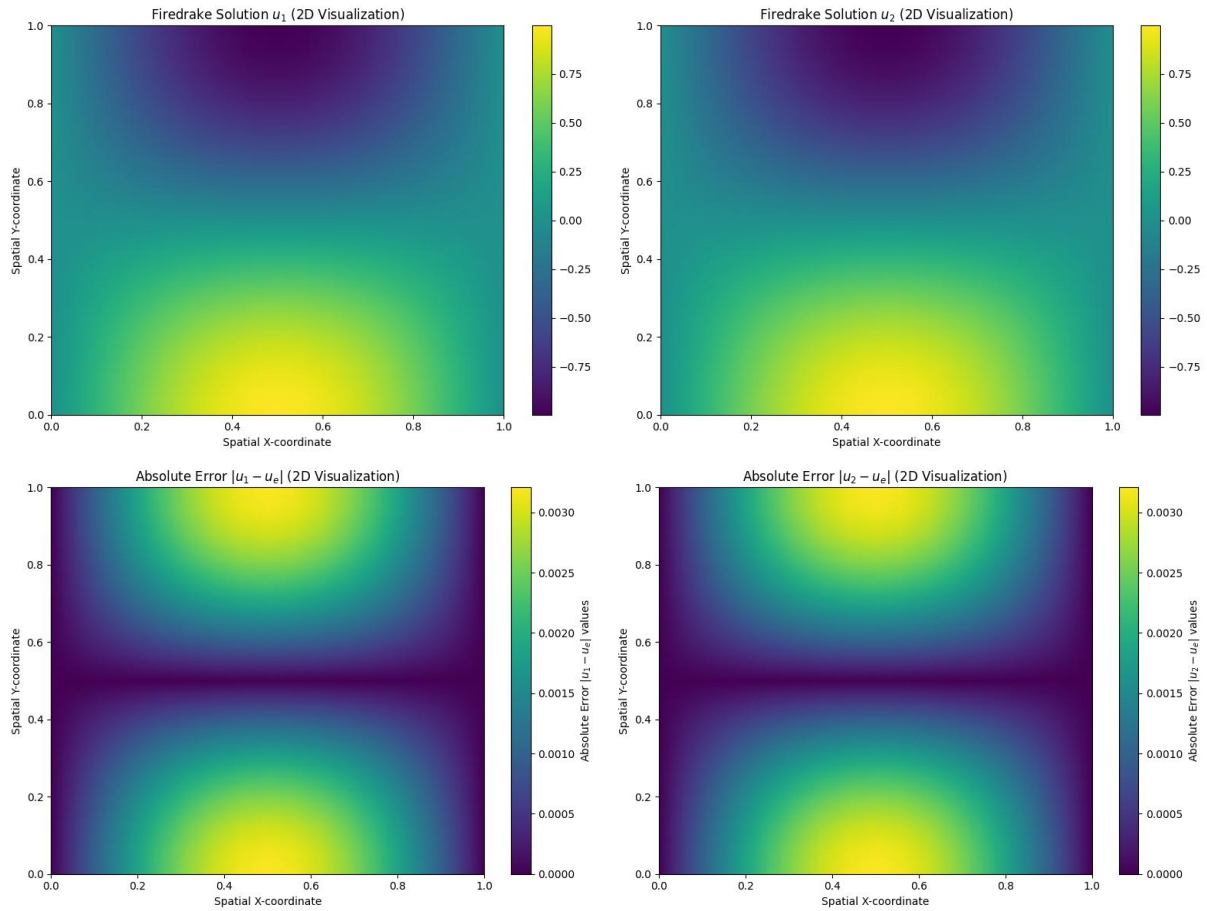


Figure 2: Solutions u_1 and u_2 obtained from Firedrake with the mesh resolution: $\Delta x = 0.0625$ and $p=1$. The absolute errors are also plotted for u_1 and u_2 .

For the mesh resolution: $\Delta x = 0.03125$ and $p=1$, the L2 error for $u_1 = 0.000400757$, and for $u_2 = 0.00040075733$.

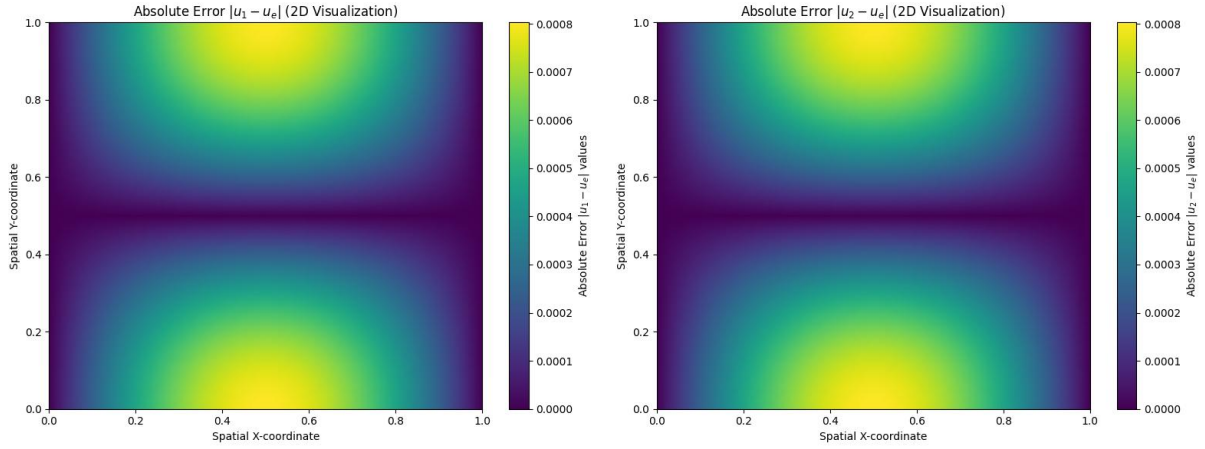


Figure 3: The absolute errors are plotted for u_1 and u_2 with the mesh resolution: $\Delta x = 0.03125$ and $p=1$.

For the mesh resolution: $\Delta x = 0.015625$ and $p=1$, the L2 error: $u_1 = 0.000100346404$, and $u_2 = 0.00010034640404$.

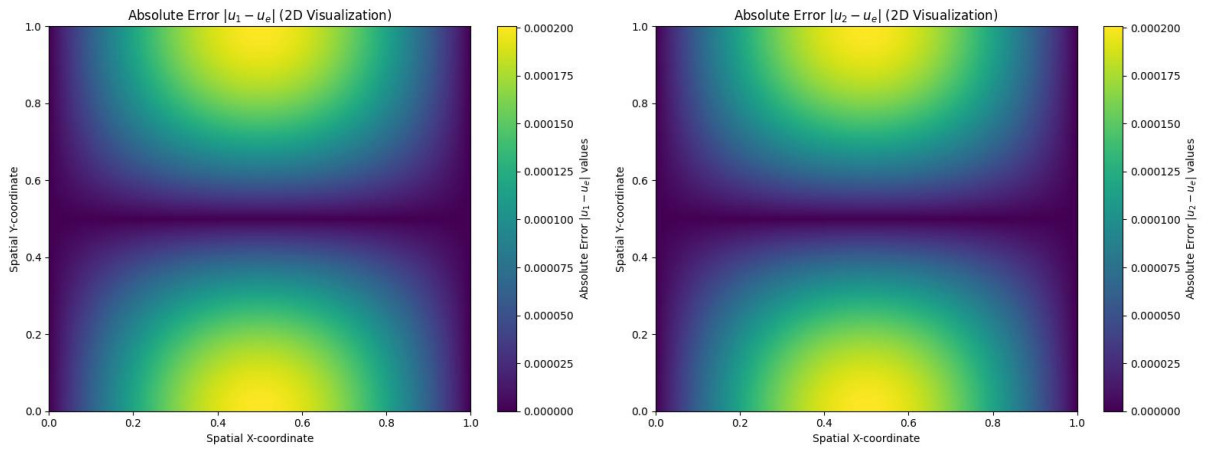


Figure 4: The absolute errors are plotted for u_1 and u_2 with the mesh resolution: $\Delta x = 0.015625$ and $p=1$

For the mesh resolution: $\Delta x = 0.0078125$ and $p=1$, the L2 error: $u_1 = 2.50964262 \times 10^{-5}$, and $u_2 = 2.50964263 \times 10^{-5}$.

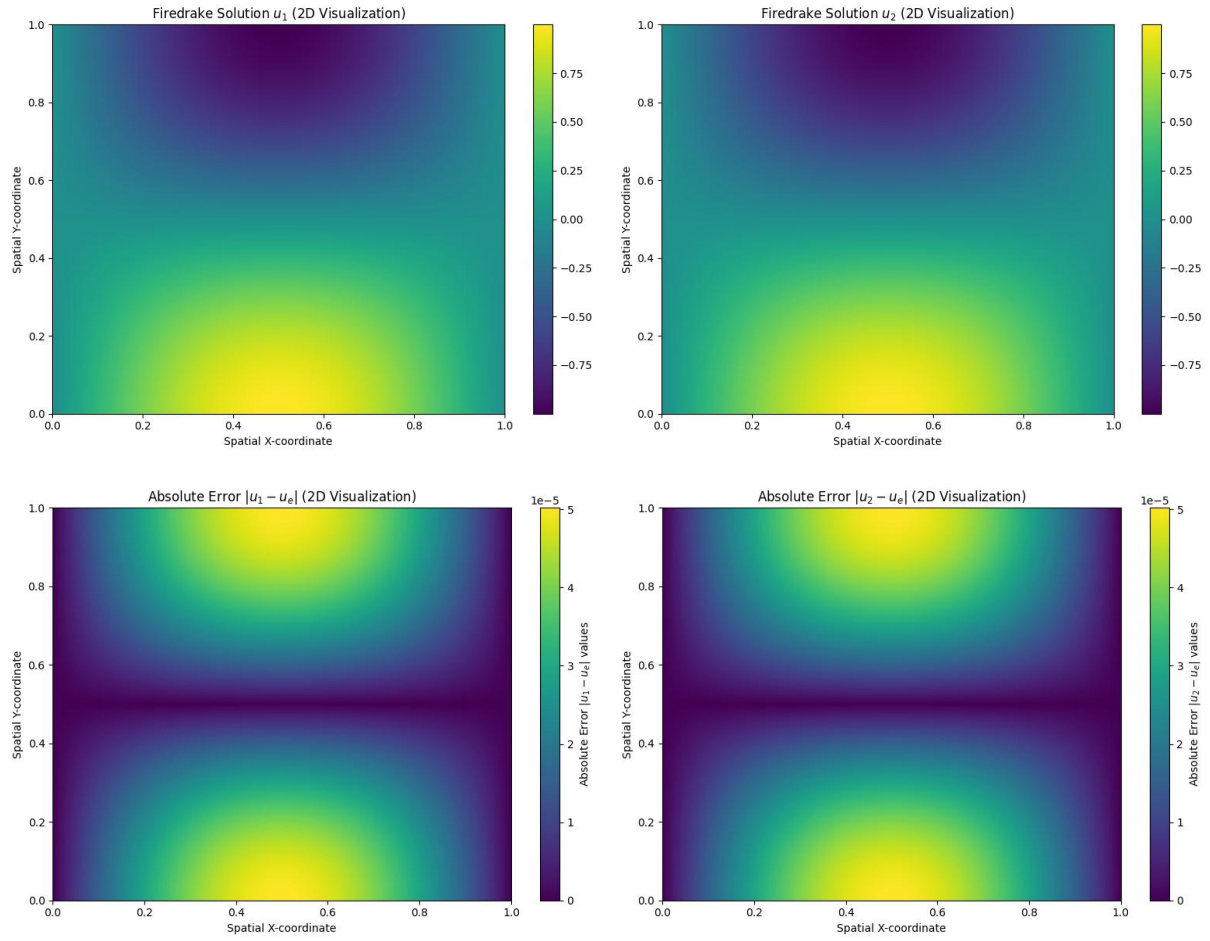
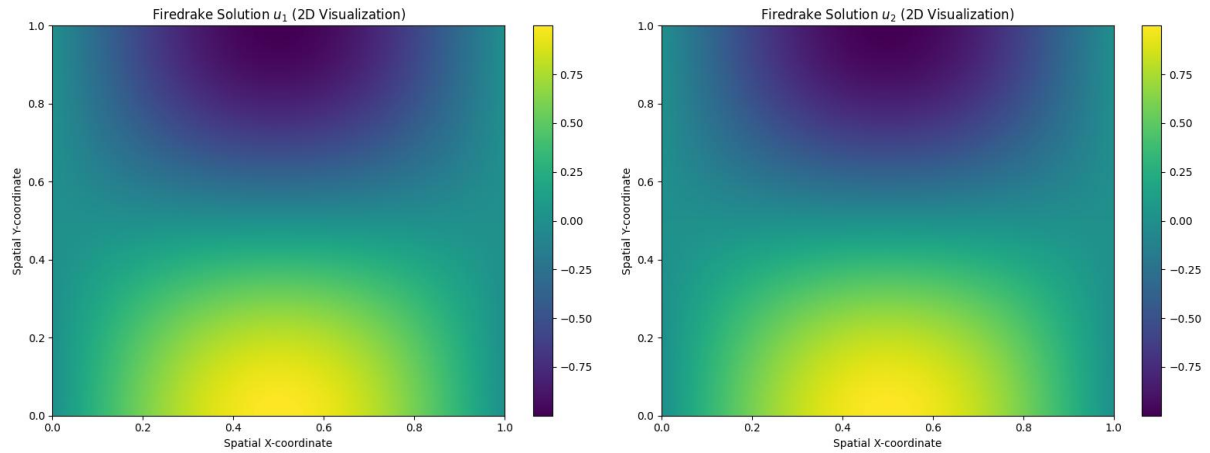


Figure 5: Solutions u_1 and u_2 obtained from Firedrake with the mesh resolution: $\Delta x = 0.0078125$ and $p=1$. The absolute errors are also plotted for u_1 and u_2 .

For mesh resolution: $\Delta x = 0.0078125$ (128x128) and $p=2$, the L2 error for u_1 is 2.555×10^{-10} , and for u_2 is 2.554×10^{-10} .



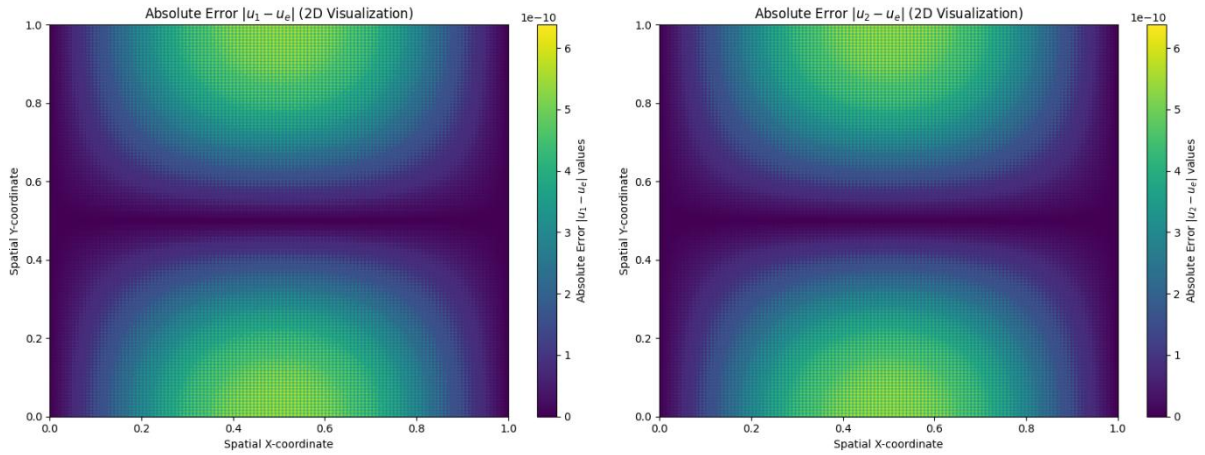


Figure 6: Solutions u_1 and u_2 for mesh size 128 and $p=2$ (top). L2 errors for these two methods (bottom).

For mesh resolution: $\Delta x = 0.0078125$ (128x128) and $p=3$, the L2 error for u_1 is 2.180×10^{-12} , and for u_2 is 2.596×10^{-12} .

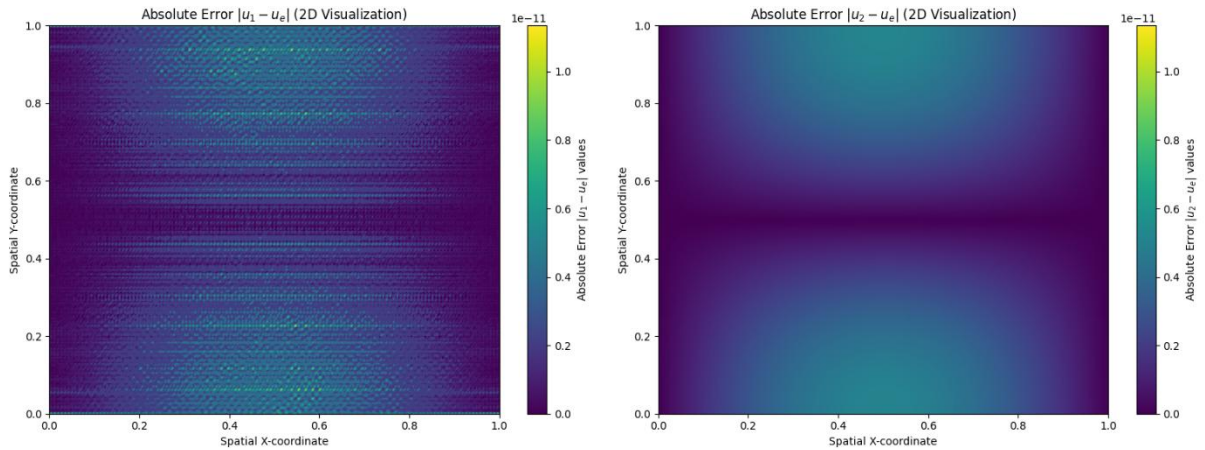


Figure 7: L2 error plots for u_1 and u_2 with mesh size 128 and $p=3$.

Figure 8 shows that as the polynomial degree increases, the convergence rate (slope of the lines) also increases. We expected an L2 error convergence rate of $O(\Delta x^{(p+1)})$ for CG_p elements. The observed average orders should closely match this theoretical prediction. For lower polynomial degrees like $p=1, 2$, the errors are larger. For higher degrees, the method is so efficient that it reaches the limits of numerical precision or solver tolerance very quickly. But, the round-off errors may also begin to dominate for higher polynomial orders with very large numbers of elements.

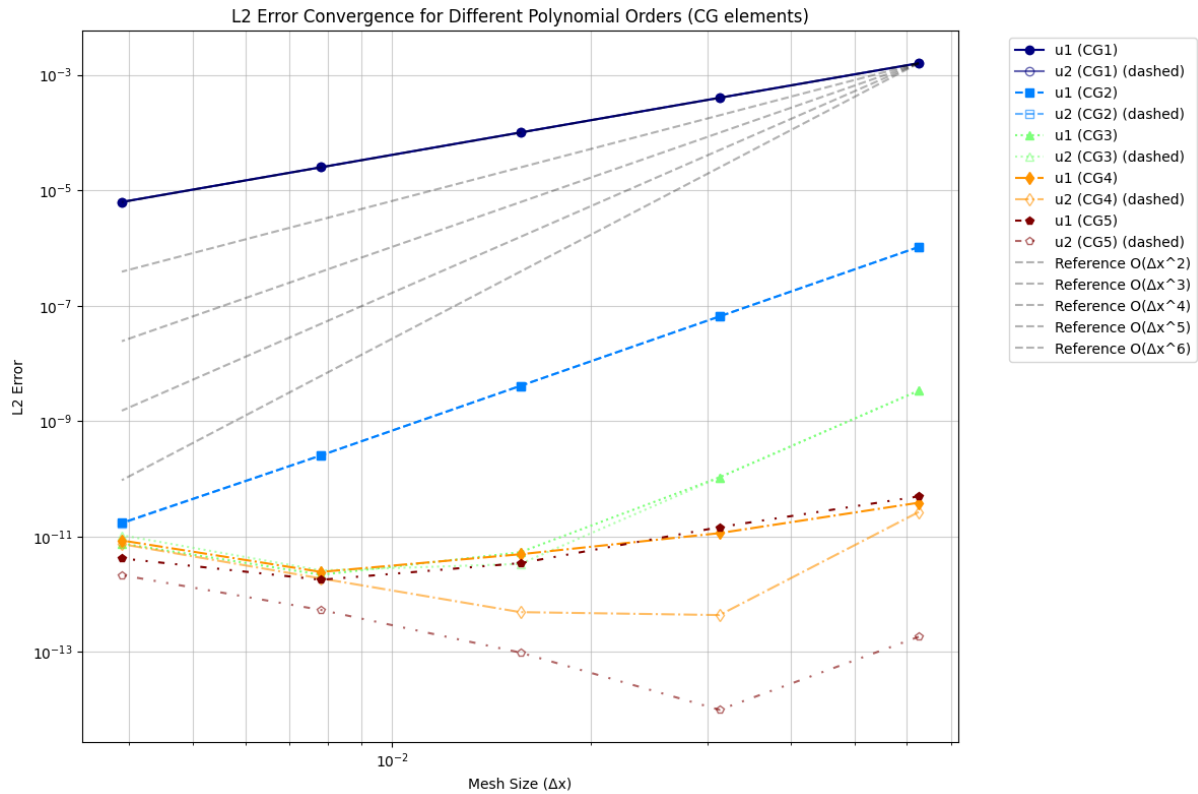


Figure 8: L2 error convergence study for solution methods u_1 and u_2 for different mesh resolutions and different polynomial orders.

In figure 9, the average observed convergence order for CG1 (u_1) is: 2.00. This result strongly aligns with the theoretical expectation of an $O(\Delta x^2)$ convergence rate for L2 error when using Continuous Galerkin elements of degree 1 for refined meshes. As Δx decreases, the error decreases quadratically, as indicated by the average order of 2.00.

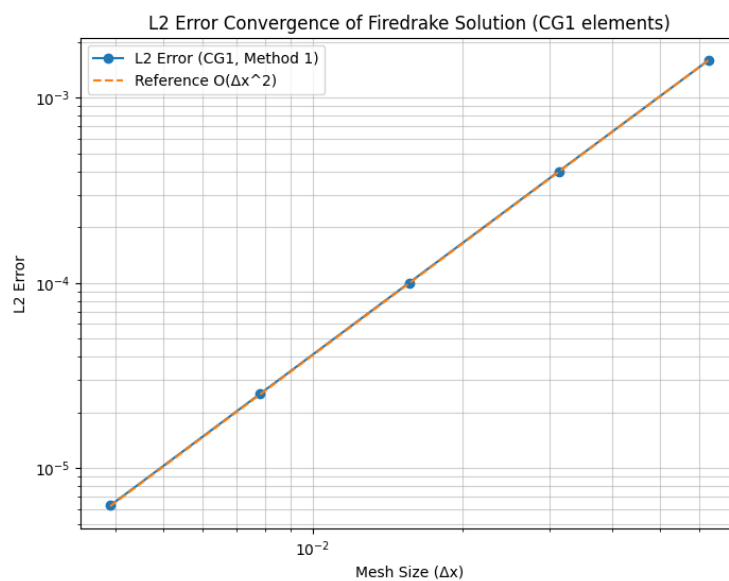


Figure 9: L2 error convergence of solution method 1 (CG1).

Question 5

The code is uploaded separately with comments and explanations!!

Question 6

For a new exact solution, $u_e = x^2 \cos(\pi y) \dots \dots \dots (29)$,

the source function f becomes, $f = (\pi^2 x)^2 \cos(\pi y) \dots \dots \dots (30)$.

Checking the Dirichlet boundary conditions:

$$u(0, y) = 0. \cos(\pi y) = 0$$

$$u(1, y) = \cos(\pi y)$$

and Neumann boundary conditions:

$$\partial_y u(x, y)|_{y=0} = -\sin(\pi \cdot 0) = 0$$

$$\partial_y u(x, y)|_{y=1} = -\sin(\pi \cdot 1) = 0$$

Figure 10 plots the numerical solution for the new source function f and the exact solution for the 2D Poisson's equation solved using the Firedrake code. The quadrilateral mesh with the mesh resolution $\Delta x = 0.0078125$ (128) is used and the degree of polynomials $p=1$ is used. The L2 Error (for plotting mesh 128x128) is calculated as 5.05×10^{-6} .

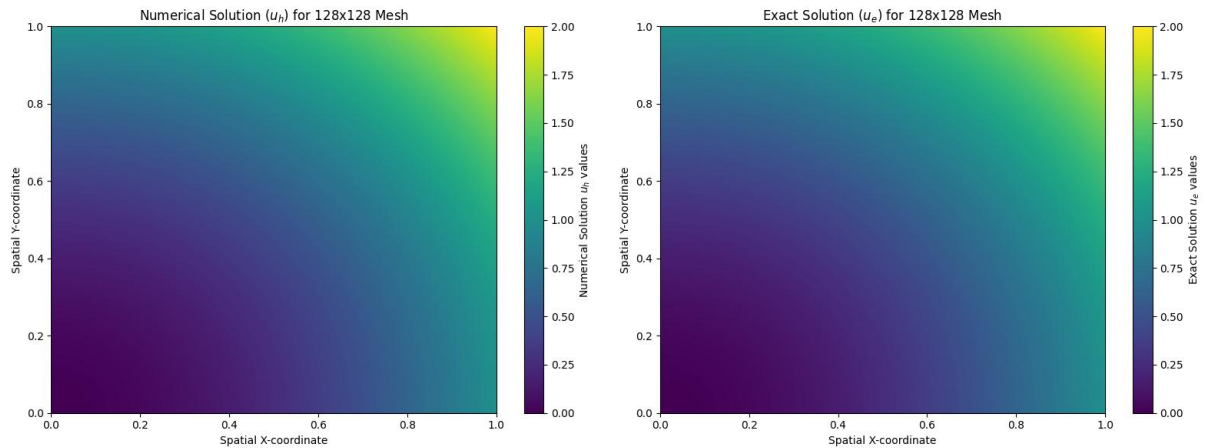


Figure 10: The numerical solution $u_h(x,y)$ and the exact solution $u_e(x,y)$ calculated for the new source function f and the exact solution u_e .

Figure 11 presents the absolute localized error field $|u_h(x,y) - u_e(x,y)|$. The solid dark purple colours along the vertical edges ($x=0$) and ($x=1$) visually confirm the Dirichlet boundary conditions. But for the horizontal edges ($y=0$ and ($y=1$), error is applied orthogonally, validating the successful implicit application of the homogeneous Neumann conditions. The discretization error bounded at a precise peak threshold of approximately 1.06×10^{-5} .

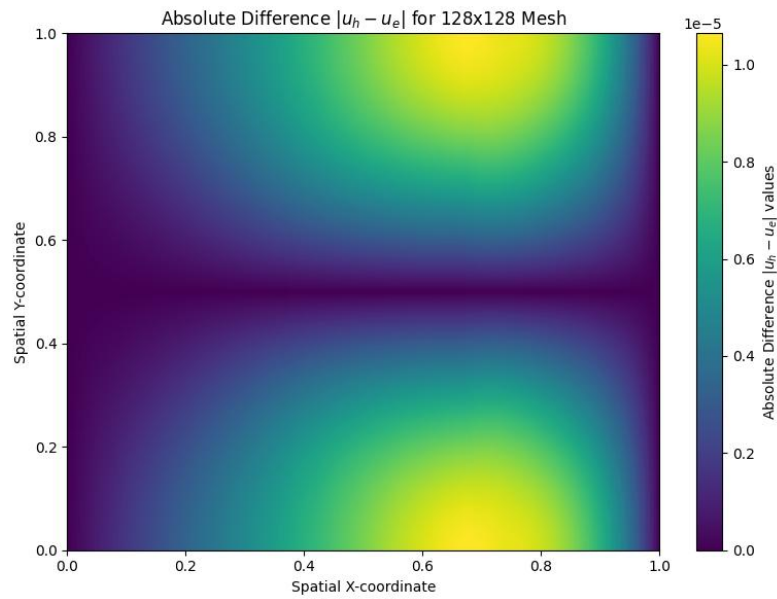


Figure 11: The absolute error $|u_h(x,y) - u_e(x,y)|$ calculated for the new source function f and the exact solution u_e .

MATH5453M Numerical Exercise 3, Part II

Ground water model and canal flow

The complete mathematical groundwater model summarized is:

$$\partial_t(\omega_v h_m) - \alpha g \partial_y(\omega_v h_m \partial_y h_m) = \frac{\omega_v R}{m_{por} \sigma_e} \text{ in } y \in [0, L_y] \dots \dots \dots (31)$$

$$\partial_y h_m = 0 \text{ at } y = L_y \text{ and } h_m(0, t) = h_{cm}(t) \text{ at } y = 0 \dots \dots (32)$$

$$L_c \omega_v \frac{dh_{cm}}{dt} = \omega_v m_{por} \frac{\sigma_e}{2} \alpha g \partial_y(h_m^2)|_{y=0} - \omega_v \sqrt{g} \max\left(\frac{2}{3} h_{cm}(t), 0\right)^{\frac{3}{2}} \dots \dots \dots (33)$$

Question 1

Discretising eq.(31)–(33) using an explicit finite element space and time discretisation:

To discretize the system, a unified trial space is introduced at $y=0$, where the groundwater height $h_m(y,t)$ and the canal height $h_{cm}(t)$ are explicitly bound $h_m(0,t) = h_{cm}(t)$. Importantly, the test function $v(y)$ can be non-zero at the boundary. So, mapping it to an arbitrary scalar test variable v_{cm} to match the canal degree of freedom is allowed $v(0)=v_{cm}$.

Now, equation (31) is multiplied by the test function $v(y)$ and is integrated over the channel entire length $\Omega=[0, L_y]$:

$$\int_0^{L_y} \partial_t (\omega_v h_m) v dy - \frac{\alpha g}{2} \int_0^{L_y} \partial_y (\omega_v \partial_y (h_m^2)) v dy = \int_0^{L_y} \frac{\omega_v R}{m_{por} \sigma_e} v dy \dots \dots (34)$$

Integral by parts,

$$\begin{aligned} \int_0^{L_y} \partial_t (h_m) v dy + \frac{\alpha g}{2} \int_0^{L_y} \partial_y v (\partial_y (h_m^2)) dy - \left[\frac{\alpha g}{2} \partial_y (h_m^2) v \right]_0^{L_y} \\ = \int_0^{L_y} \frac{R}{m_{por} \sigma_e} v dy \dots \dots (35) \end{aligned}$$

The upper boundary term vanishes due to the other boundary condition, $\partial_y h_m = 0$ at $y = L_y$.

$$\begin{aligned} \int_0^{L_y} \partial_t (h_m) v dy + \frac{\alpha g}{2} \int_0^{L_y} \partial_y v (\partial_y (h_m^2)) dy + \frac{\alpha g}{2} \partial_y (h_m^2) \Big|_{y=0} v(0) \\ = \int_0^{L_y} \frac{R}{m_{por} \sigma_e} v dy \dots \dots (36) \end{aligned}$$

$$\begin{aligned} m_{por} \sigma_e \int_0^{L_y} \partial_t (h_m) v dy + m_{por} \sigma_e \frac{\alpha g}{2} \int_0^{L_y} \partial_y v (\partial_y (h_m^2)) dy + m_{por} \sigma_e \frac{\alpha g}{2} \partial_y (h_m^2) \Big|_{y=0} v(0) \\ = \int_0^{L_y} R v dy \dots \dots \dots (37) \end{aligned}$$

Applying the boundary conditions and eq. (33),

$$\begin{aligned} m_{por} \sigma_e \int_0^{L_y} \partial_t (h_m) v dy + m_{por} \sigma_e \frac{\alpha g}{2} \int_0^{L_y} \partial_y v (\partial_y (h_m^2)) dy + L_c \omega_v \frac{dh_{cm}}{dt} v_{cm} \\ + \omega_v \sqrt{g} \max \left(\frac{2}{3} h_{cm}(t), 0 \right)^{\frac{3}{2}} v_{cm} = \int_0^{L_y} R v dy \dots \dots \dots (38) \end{aligned}$$

This is the monolithic weak formulation.

I think, the condition $h_m(0,t) = h_{cm}(t)$ can be enforced strongly within the trial function expansion but for the Dirichlet constraint on the test function space, it might force $v(0)=0$.

For the spatial Finite Element discretization, approximating the continuous groundwater height using standard piecewise linear shape functions $\phi_i(y)$ across the mesh elements:

$$h_m(y, t) = \sum_i h_i(t) \phi_i(y) \dots \dots \dots (39)$$

And the boundary node coefficient matches the canal state: $h_0(t) = h_{cm}(t)$.

Forward Euler or explicit formulation sets the time-stepping parameter $\theta = 0$. All non-linear spatial diffusion and weir sink terms are evaluated entirely at known time level n (also current), but the transient time derivatives to $n+1$ (advancing):

$$\begin{aligned} m_{por} \sigma_e \int_0^{L_y} \left(\frac{h_m^{n+1} - h_m^n}{\Delta t} \right) \phi_i dy + m_{por} \sigma_e \frac{\alpha g}{2} \int_0^{L_y} \partial_y \phi_i (\partial_y (h_m^n)^2) dy \\ + L_c \left(\frac{h_{cm}^{n+1} - h_{cm}^n}{\Delta t} \right) \delta_{i,0} + \sqrt{g} \max \left(\frac{2}{3} h_{cm}^n, 0 \right)^{\frac{3}{2}} \delta_{i,0} \\ = \int_0^{L_y} R_n \phi_i dy \dots \dots \dots (40) \end{aligned}$$

Then the explicit mass matrix becomes,

$$\mathbf{M} \left(\frac{h_m^{n+1} - h_m^n}{\Delta t} \right) = \mathbf{F}^n \dots \dots \dots (41),$$

And the estimate of the explicit time step stability is,

$$\Delta t \leq \frac{\Delta y^2}{2\alpha g \max(h_m^n)} \dots \dots \dots (42).$$

Question 2

The parameter values given for the numerical simulations:

$$m_{por} = 0.3, \sigma_e = 0.8, L_y = 0.85\text{m}, k = 10^{-8} \text{ m}^2, \omega_v = 0.1\text{m}, R_{max} = 0.000125\text{m/s}, L_c = 0.05\text{m}.$$

These simulations are run using $m = 32$ for high resolution, $\theta = 0$, and $P = 1$.

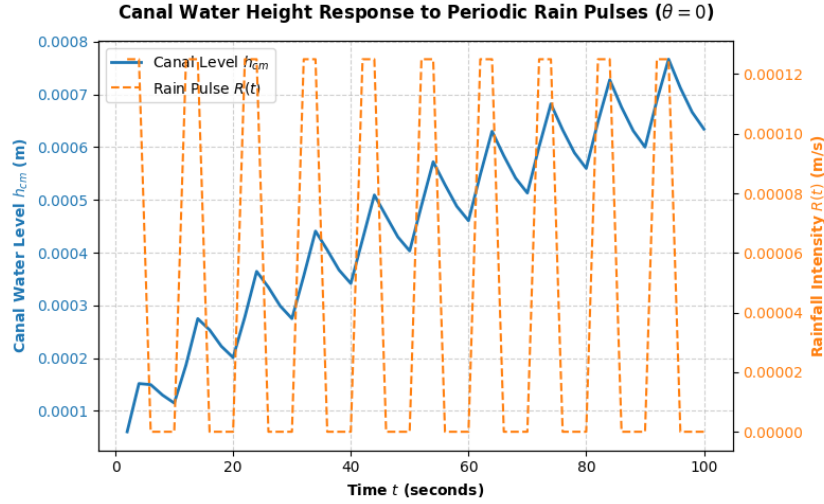


Figure 12: Highly transient, jagged sawtooth oscillation of the canal water height h_{cm} as it responds dynamically to the 10-second periodic rainfall pulses.

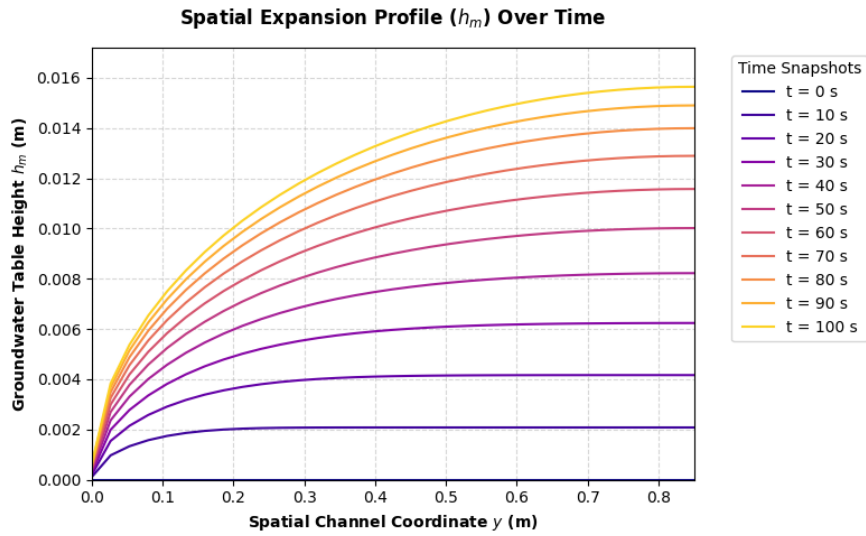


Figure 13: The spatial evolution of the groundwater table h_m using 10-second snapshots ($\theta=0$ & $P=1$).

During the run it is found that the Range of last 10 h_{cm} values: 1.665674×10^{-04} m and the Standard deviation of last 10 h_{cm} values: 4.771665×10^{-05} m. Therefore, it is concluded that the canal height (h_{cm}) has not yet reached a steady state based on the last 10 values from this simulation. Additionally, the last two spatial profiles ($t=90.00$ s and $t=100.00$ s): the L2-norm difference between last two profiles is 3.351892×10^{-03} . So, the spatial profile ($h_m(y)$) has not reached a steady state situation either.

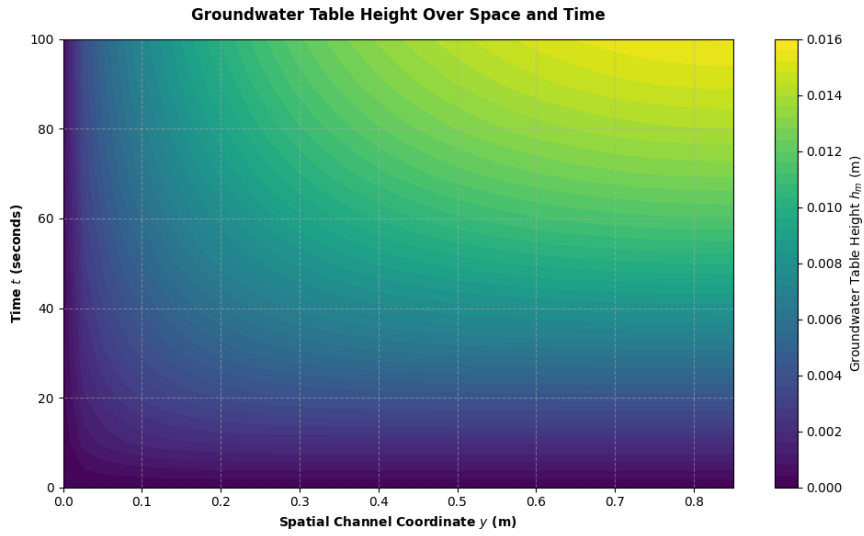


Figure 14: A continuous 2D space-time contour map of the entire system showing initial sharp infiltration front pushing through the dry soil during early times.

To make a comparison between the duration of different pulses, the simulations were run using $m=16$ only to make it faster. All other parameters were kept identical.

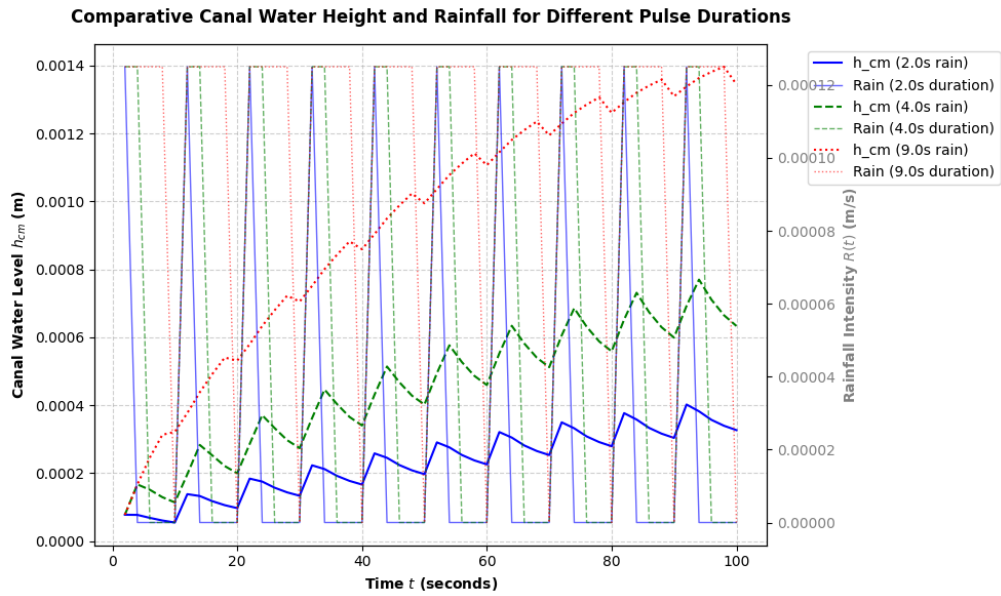


Figure 15: For 2.0s, 4.0s, and 9.0s durations within a 10-second cycle, comparison of the canal water height h_{cm} alongside the corresponding periodic rainfall pulse tracks.

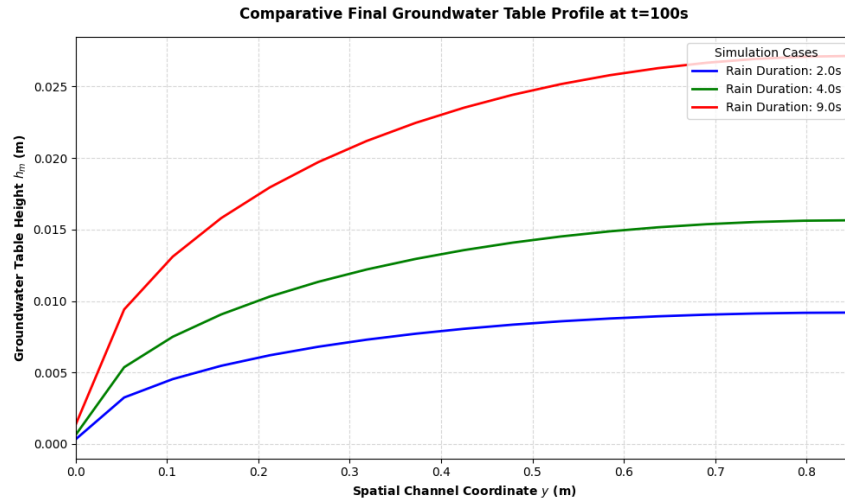


Figure 16: A comparison of the final groundwater table height h_m profiles at $t = 100s$ across the entire channel length for durations 2.0s, 4.0s, and 9.0s rain pulse conditions.

Question 3

In figure 17, the global solution envelope matches the explicit benchmark. It reaches a transient peak of 0.00076m at $t = 100s$. At the sharpest sawtooth tips, a slight rounding reflects the mathematical time-averaging of the forcing terms over the discrete interval. The simulation run much faster than the ($\theta = 0.0$) and $P=1$ cases.

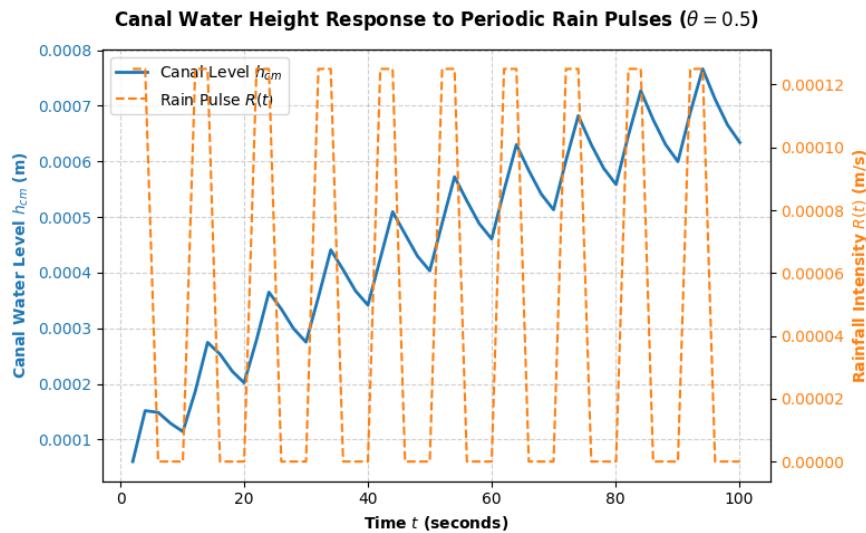


Figure 17: The canal water level response (h_{cm}) under the semi-implicit Crank-Nicolson formulation ($\theta = 0.5$) with a step size of ($Dt = 0.1$) and $P=1$.

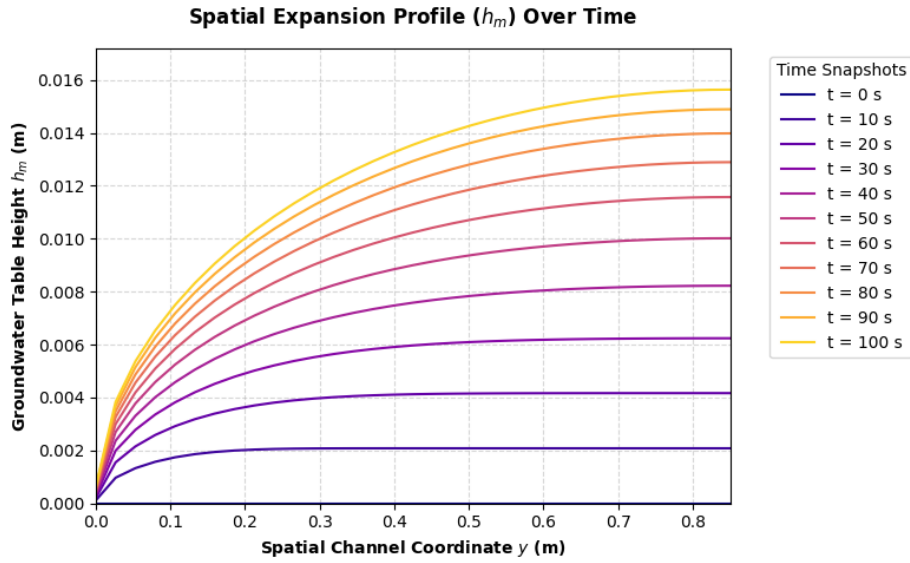


Figure 18: The spatial evolution snapshots of the groundwater table height ($h_m(y)$) calculated using the Crank-Nicolson scheme ($\theta = 0.5$, $P=1$).

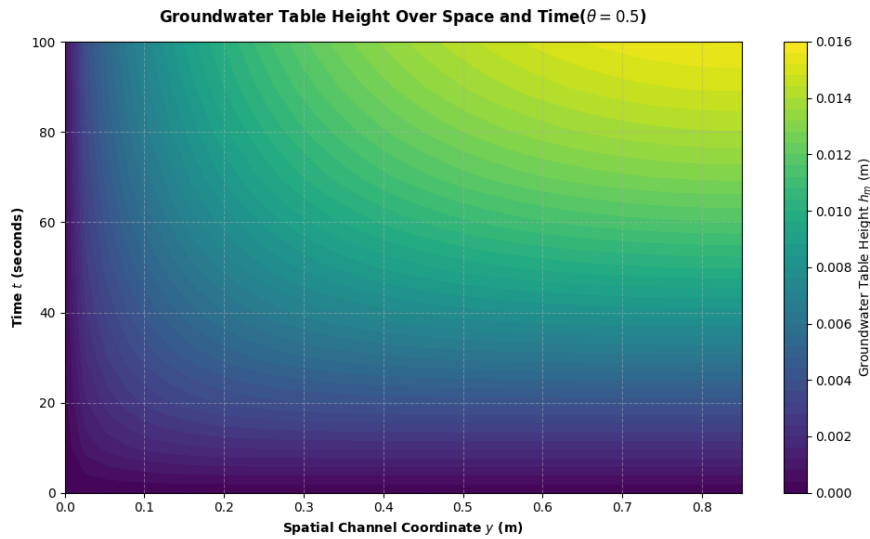


Figure 19: A continuous 2D space-time contour map tracking the groundwater table height $h_m(y,t)$ calculated using the Crank-Nicolson scheme ($\theta = 0.5$, $P=1$).

Question 4

From figure 20, the clear structural failure of higher-order continuous formulations is seen when subjected to non-smooth, sharp periodic rainfall inputs. Additionally, figure 21 also exposes the local spatial failure of higher-order continuous formulations near the boundary interface, $y \leq 0.05$.

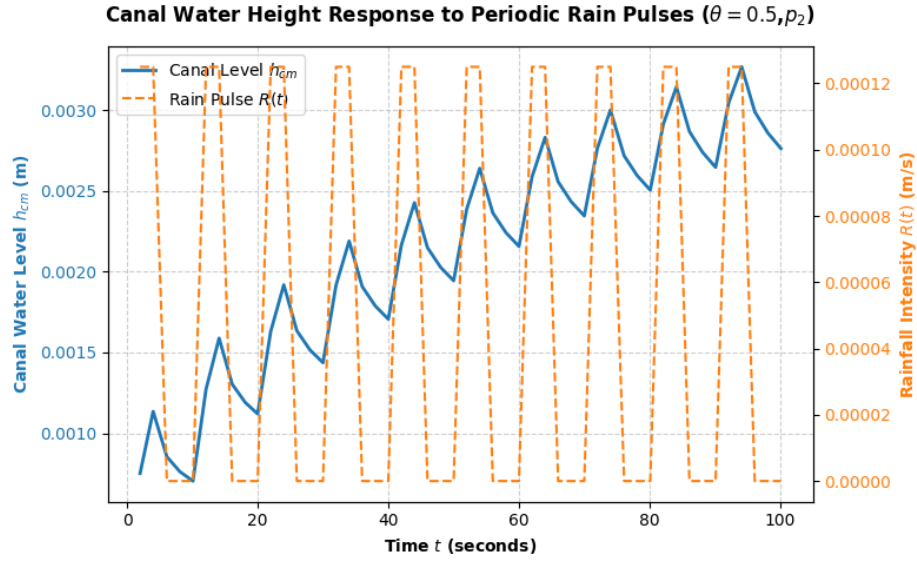


Figure 20: The transient response of the canal water height h_{cm} using quadratic polynomial elements (CG_2) under a semi-implicit Crank-Nicolson framework ($\theta = 0.5$).

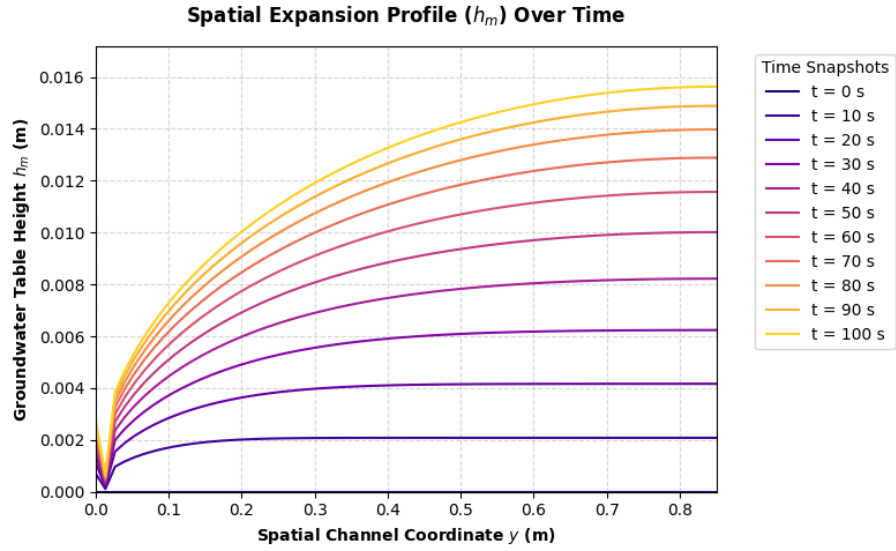


Figure 21: The spatial evolution profiles of the groundwater table h_m tracked at 10-second intervals under the continuous quadratic polynomial space (CG_2).